

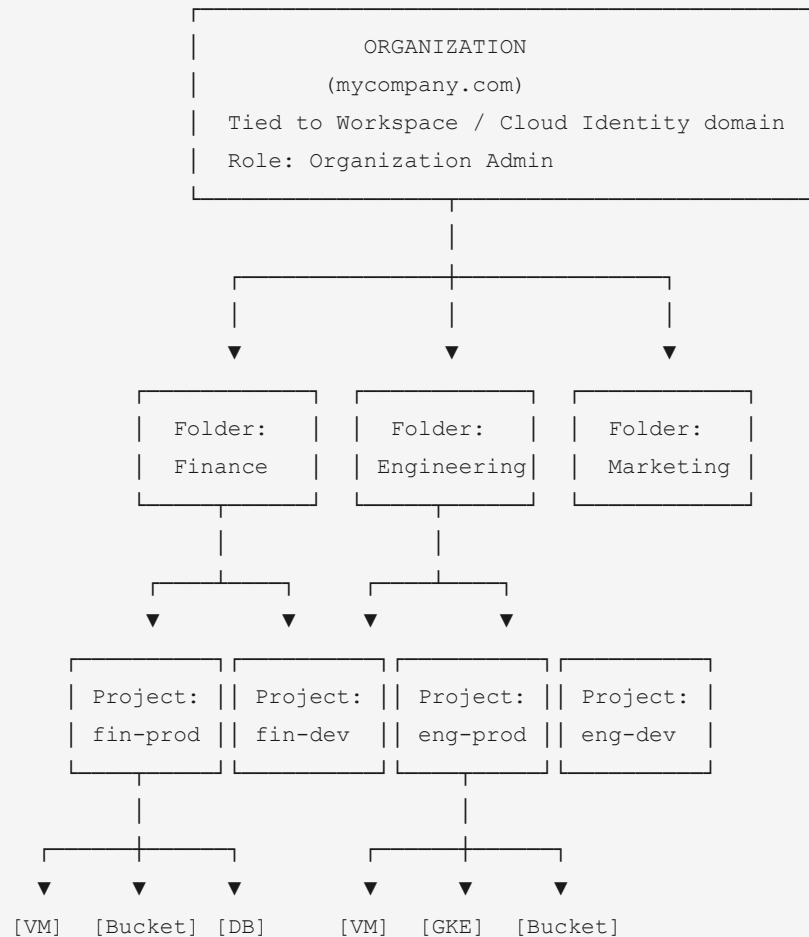
GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 1 -- Visual Maps: Setting Up a Cloud Solution Environment

GCP Study Notes

Study Notes April 2026

1. Google Cloud Resource Hierarchy



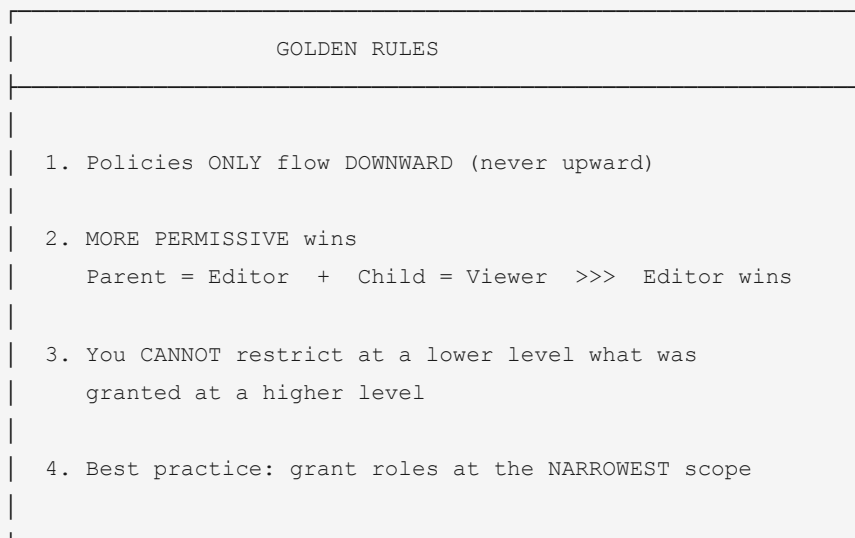
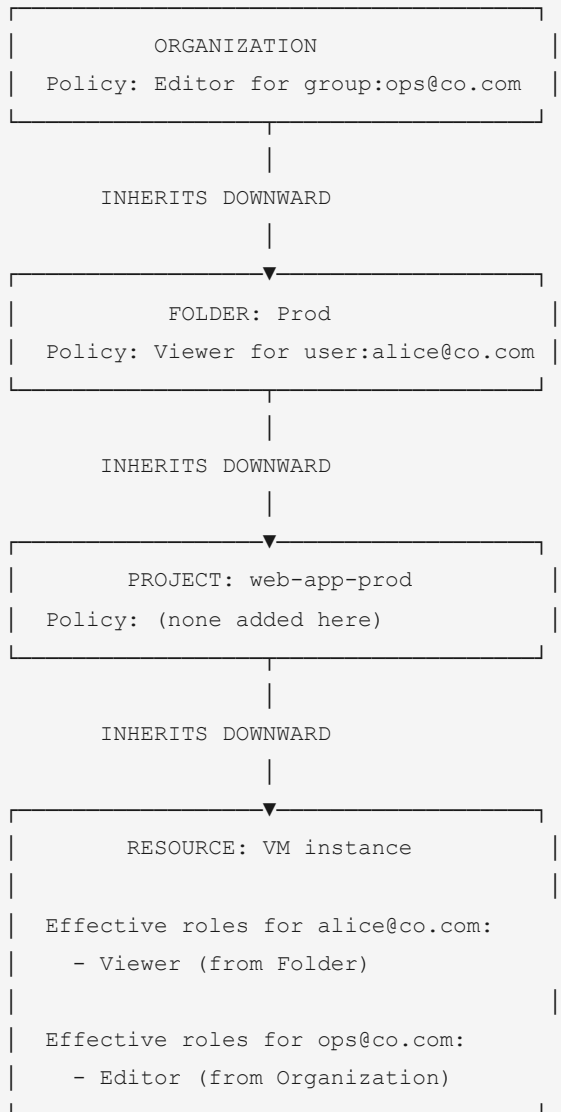
Key Rules

HIERARCHY RULES		
1. Organization	— optional but recommended (needs Workspace or Cloud Identity)	
2. Folders	— optional, up to 10 levels deep	
3. Projects	— MANDATORY (every resource needs one)	
4. Resources	— always belong to exactly ONE project	

Project Identifiers

PROJECT IDENTIFIERS		
Identifier	Unique?	Who Sets It?
Project Name	NOT unique	You choose, can change later
Project ID	GLOBALLY unique	You choose at creation, FIXED
Project Num	GLOBALLY unique	Auto-assigned by Google, FIXED

2. IAM Inheritance Diagram



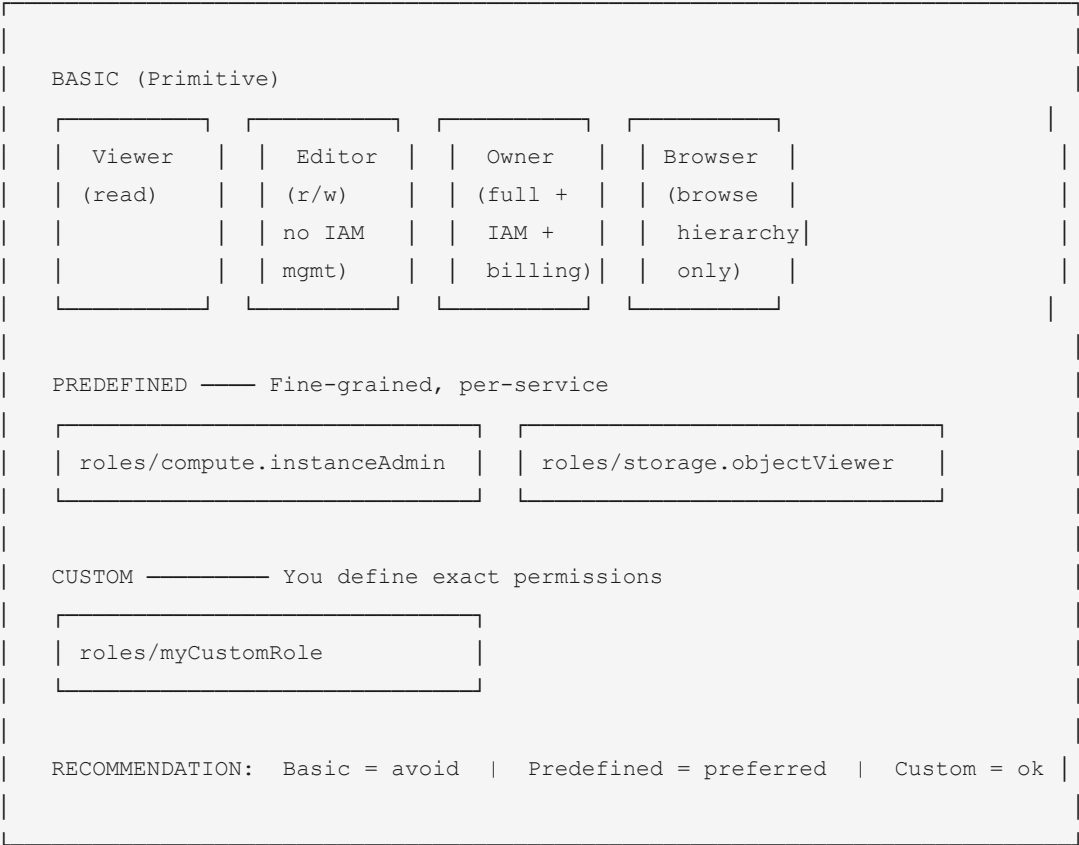
3. IAM Model – Who / What / Which

WHO (Member/Principal)	WHAT (Role)	WHICH (Resource)
user:alice@co.com	+ roles/viewer	+ project:my-app
(Identity)	(Permissions)	(Scope)

Member Types

Members / Principals	—	user:alice@co.com	Individual Google Account
	—	serviceAccount:sa@proj.iam ...	Application identity
	—	group:devs@co.com	Google Group (BEST PRACTICE)
	—	domain:company.com	Entire Workspace domain
	—	allAuthenticatedUsers	Any Google account
	—	allUsers	Anyone on internet (DANGER)

Role Types



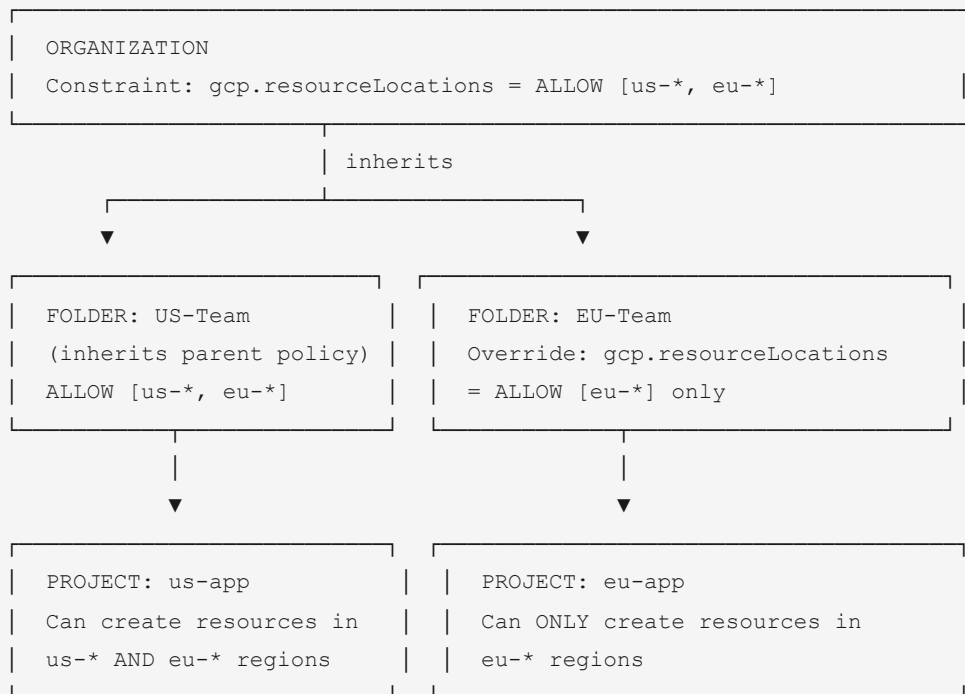
4. Organization Policy vs IAM – Comparison

IAM	ORGANIZATION POLICIES
Controls WHO can do things	Controls WHAT can be done
Grants roles to members	Sets constraints on resources
"Alice can create VMs"	"No VM may have external IP"
Applied to Org/Folder/Project	Applied to Org/Folder/Project
Inherits downward	Inherits downward
More permissive wins	Most restrictive wins

Common Org Policy Constraints Mind Map

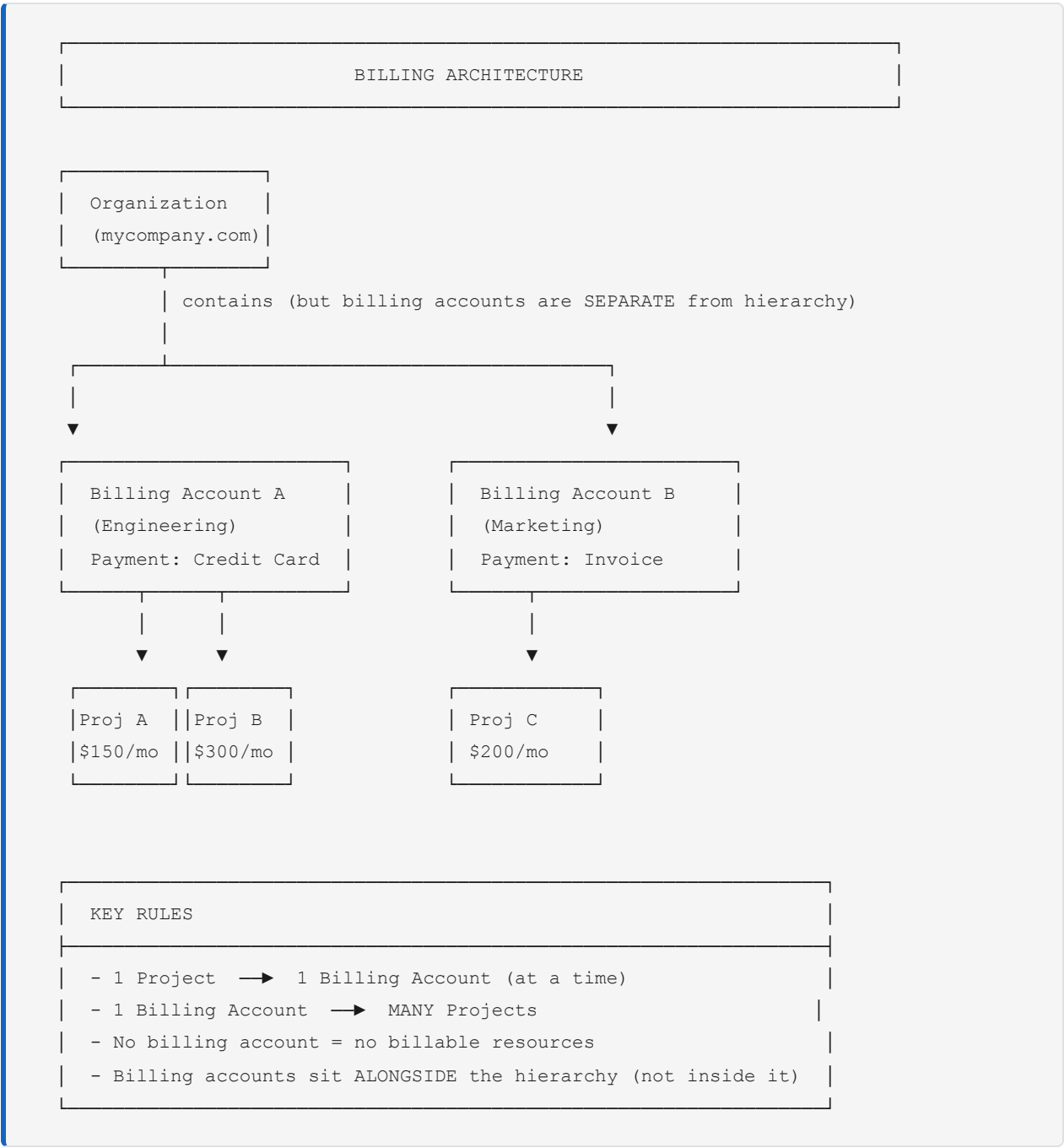


5. Policy Inheritance Flow – Org Policies



NOTE: Org policies can be OVERRIDDEN at lower levels (unlike IAM where more permissive always wins). Org policies can be set to DENY or ALLOW lists, and child nodes can further restrict.

6. Billing Flow – How Money Moves



7. Budget Alerts – Process Flow

Step 1
CREATE BUDGET

Scope:
- Billing account
- Project
- Service
- Labels

Step 2
SET THRESHOLDS

50% actual
90% actual
100% actual
100% forec. (custom %)

Step 3
CHOOSE ACTION

Email admins
Monitoring channels
Pub/Sub

Step 4
OPTIONAL: AUTO-RESPOND

Budget Alert
▼
Pub/Sub Topic
▼
Cloud Function
▼
Disable billing
Scale down resources

WARNING: Budgets DO NOT automatically stop spending!
They ONLY send notifications. You need Pub/Sub + Cloud Function to programmatically cap costs.

8. Billing Export – Decision Flow

Need to analyze billing data?



WHERE should billing data go?



BigQuery (RECOMMENDED)

- Near real-time
- Line-item detail
- SQL queries
- Free to export
- 3 types:
 1. Standard usage
 2. Detailed usage
 3. Pricing data

Cloud Storage (LEGACY)

- Daily CSV/JSON files
- Less detailed
- Good for external tools
- Being deprecated

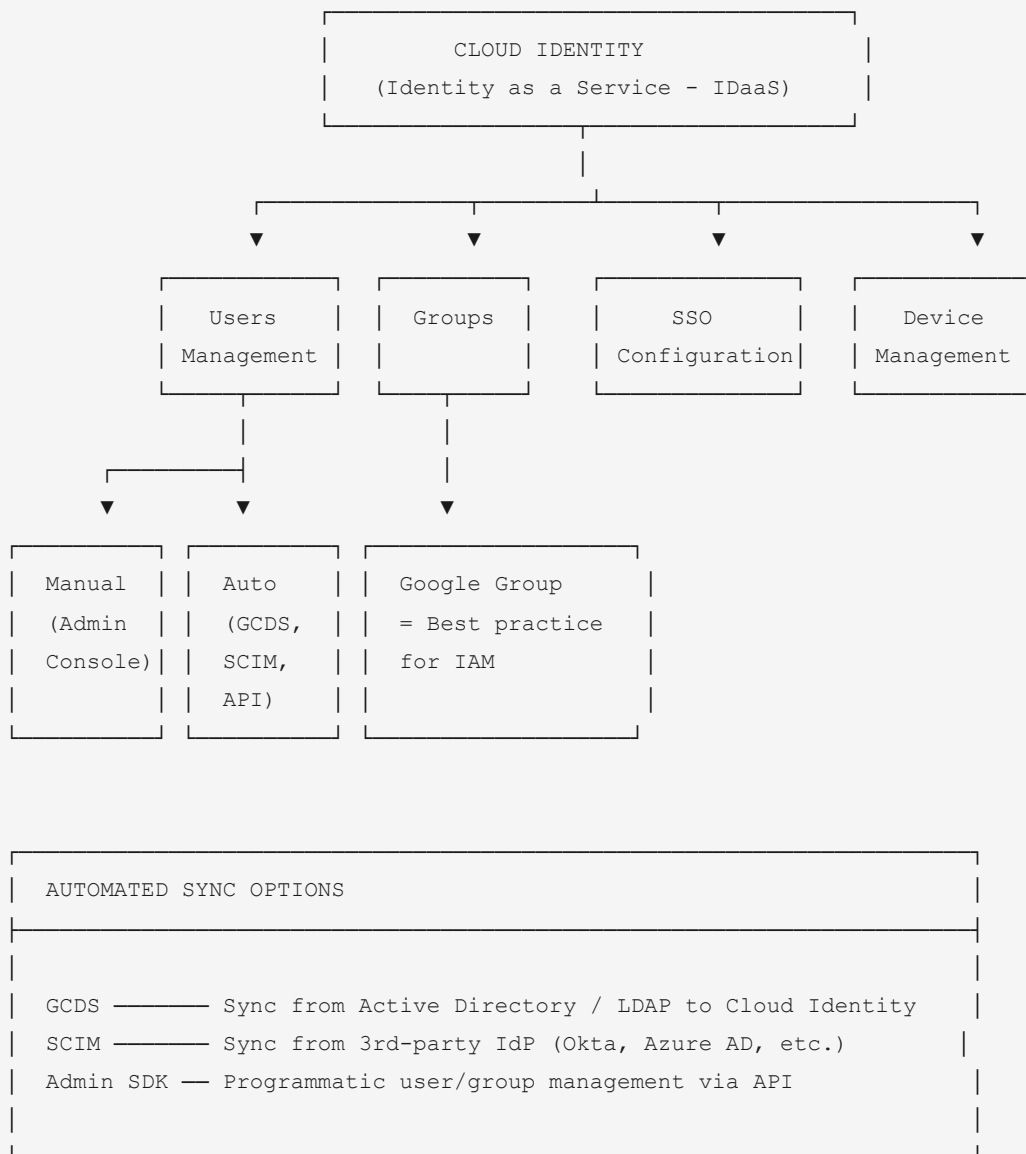
9. Billing Roles – Comparison Matrix

Capability	creator	admin	user	viewer	proj- Manager	costs- Manager
Create billing accts	X					
Manage payments		X				
Link/unlink projects		X	X		X	
View costs		X		X		X
Manage budgets		X				X
Export billing data		X				X

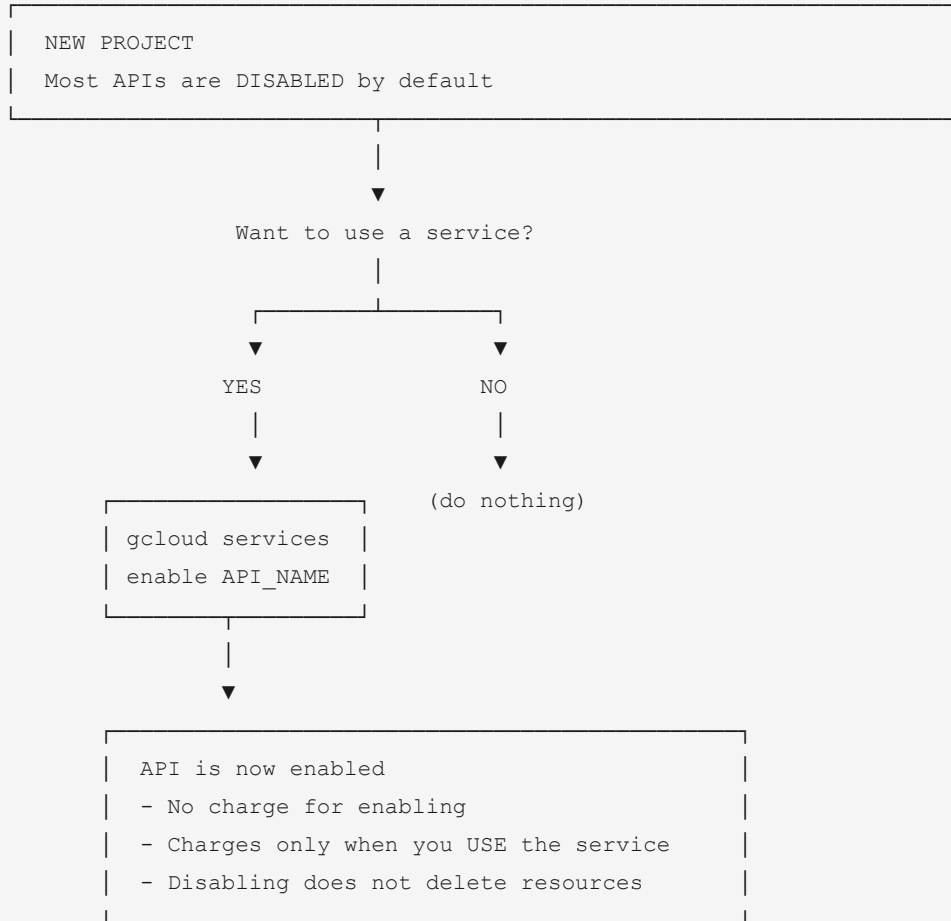
Roles:

creator = roles/billing.creator
 admin = roles/billing.admin
 user = roles/billing.user
 viewer = roles/billing.viewer
 projManager = roles/billing.projectManager
 costsManager = roles/billing.costsManager

10. Cloud Identity and User Management



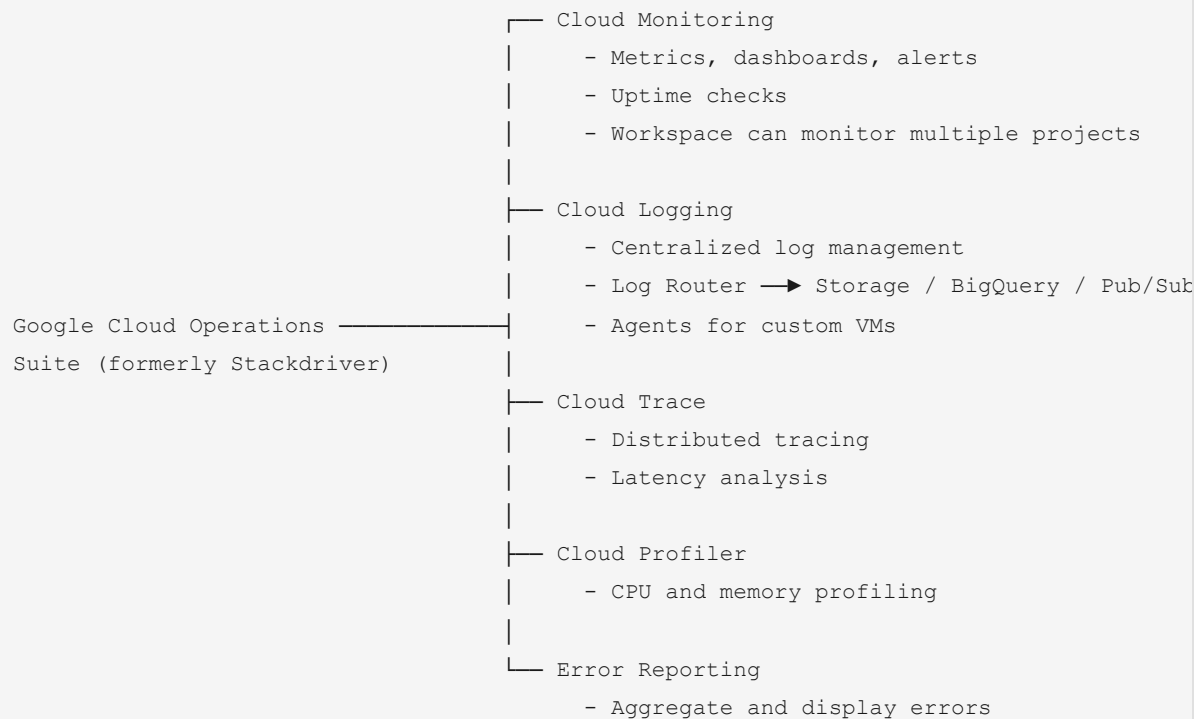
11. APIs – Enable/Disable Process



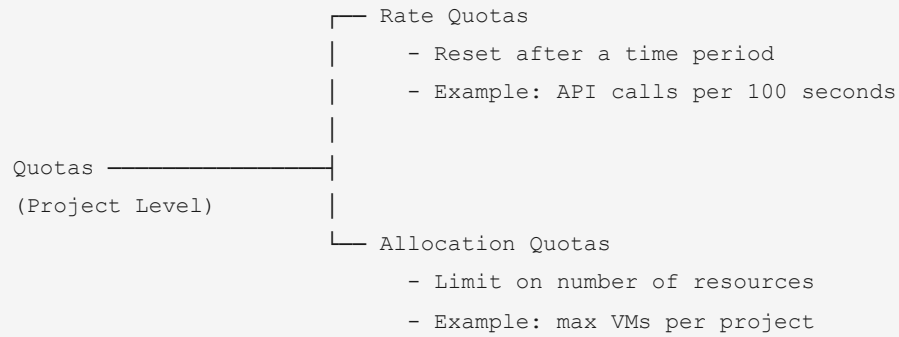
COMMONLY NEEDED APIs:

API Name	Service
compute.googleapis.com	Compute Engine
container.googleapis.com	GKE
storage.googleapis.com	Cloud Storage
sqladmin.googleapis.com	Cloud SQL
bigquery.googleapis.com	BigQuery
run.googleapis.com	Cloud Run
cloudfunctions.googleapis.com	Cloud Functions
monitoring.googleapis.com	Cloud Monitoring
logging.googleapis.com	Cloud Logging

12. Operations Suite Mind Map



13. Quotas Mind Map



QUOTA INCREASE PROCESS

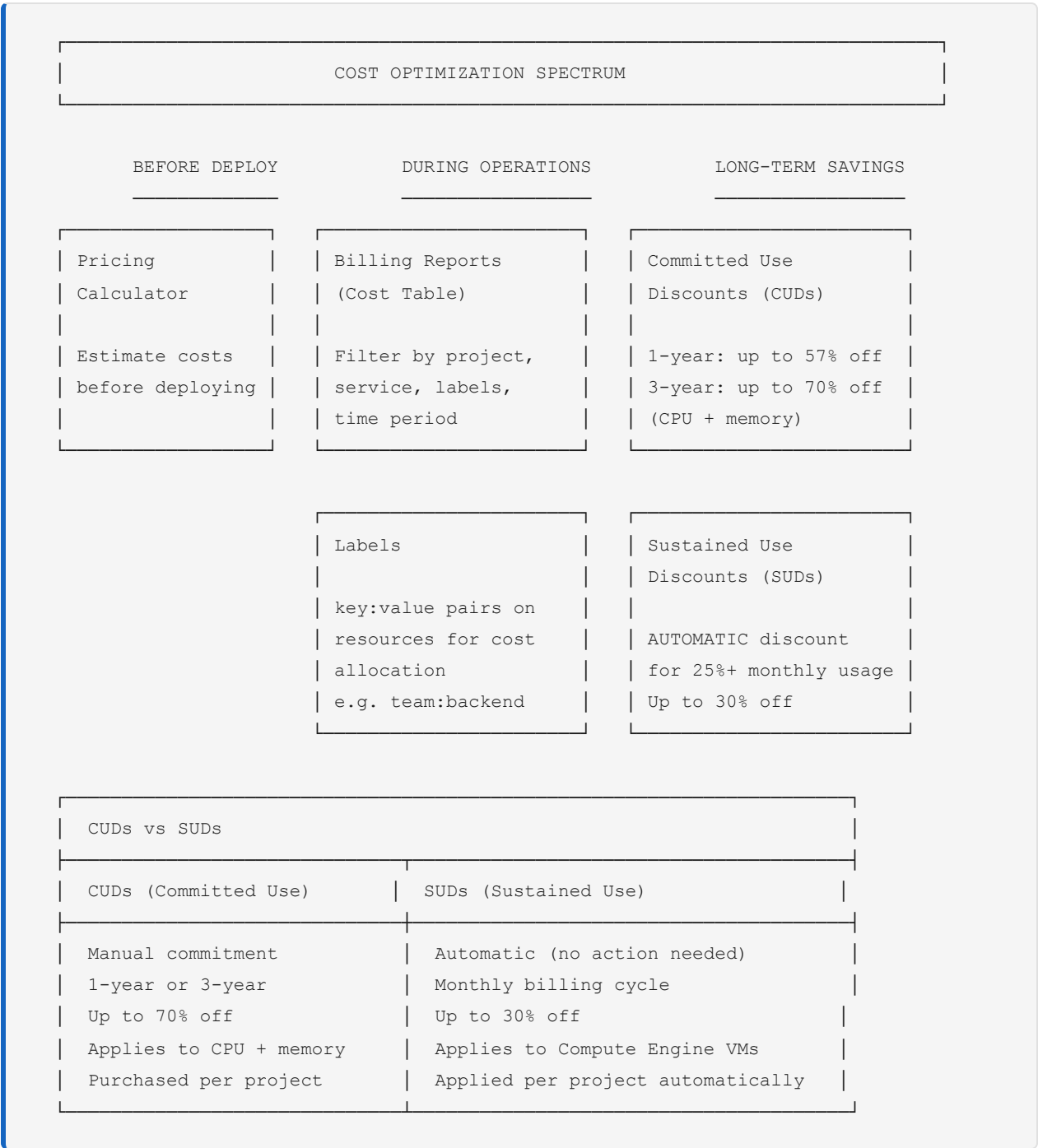
1. IAM & Admin → Quotas page
2. Filter by service / metric / location
3. Select quota → Edit Quotas
4. Enter new limit + justification
5. Submit → Google reviews (24-48 hours)

EXAM TIP: Quota errors = HTTP 429 (Too Many Requests)

EXAM TIP: Some quotas are HARD LIMITS (cannot be increased)

EXAM TIP: Monitor quota usage via Cloud Monitoring

14. Cost Management Tools Overview



15. Complete Section 1 Exam Cheat Sheet

SECTION 1: QUICK REFERENCE

HIERARCHY: Org → Folders (optional) → Projects (required) → Res.

IAM: WHO + WHAT ROLE + WHICH RESOURCE

INHERITANCE: Flows DOWN, more permissive wins

ORG POLICIES: Control WHAT can be done (constraints)

BILLING: Separate from hierarchy, 1 project = 1 billing account

BUDGETS: Alert only, do NOT cap spending

EXPORT: BigQuery (recommended) or Cloud Storage (legacy)

QUOTAS: Project-level, HTTP 429 on exceed

APIs: Disabled by default, enable before use

GROUPS: Best practice for IAM at scale

CLOUD ID: IDaaS, sync with GCDS/SCIM